

30/4/26

ASSESSMENT TOOL-2

SHORT ANSWER TEST

1. For the grammar

$E \rightarrow E + T \mid T$ remove left recursion

$$A \rightarrow A\alpha \mid B$$

$$A \rightarrow \beta A'$$

$$A' \rightarrow \alpha A' \mid \epsilon$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

2. Find $\text{FIRST}(E)$ for:

$$E \rightarrow TE', E' \rightarrow +TE' \mid \epsilon$$

Given

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$\text{FIRST}(E) = \text{FIRST}(T)$$

$$T \rightarrow id$$

$$\text{FIRST}(E) = \{id\}$$

3. For the grammar $E \rightarrow T + E \mid T$

ambiguous or not

$$E \rightarrow T + E \mid T$$

$$T \rightarrow T + T$$

$$E \Rightarrow T + E \Rightarrow T + T + E \Rightarrow T + T + T$$

The grammar is unambiguous

4. Identify whether string aba is valid for:

$$S \rightarrow aSa/b$$

$$S \rightarrow aSa$$

$$\Rightarrow aba$$

' aba ' is valid.

5. Check whether the string ab is generated by:

$$S \rightarrow aA, A \rightarrow b$$

$$S \rightarrow aA$$

$$A \rightarrow b$$

$$S \Rightarrow aA$$

$$\Rightarrow ab$$

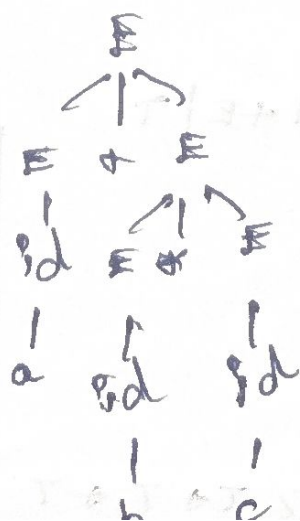
" ab " is generated

6. Construct whether the string ab is parse tree for $a+b*c$ using

$$E \rightarrow E+E | E * E | id$$

$$a+(b*c)$$

Parse Tree



$$a+(b*c)$$

7. Apply left factoring for:

$$S \rightarrow aA | aB | b$$

The common prefix is a.

$$S \rightarrow aA | aB | b$$

$$S \rightarrow aS' | b$$

$$S' \rightarrow A | B$$

8. Find FOLLOW (A) for:

$$S \rightarrow AB, B \rightarrow b$$

$$S \rightarrow AB$$

$$B \rightarrow b$$

$$\text{FOLLOW}(A) = \text{FIRST}(B)$$

$$\text{FIRST}(B) = \{b\}$$

$$\text{FOLLOW}(A) = \{b\}$$

9. Given the grammar

$$S \rightarrow aSb | \epsilon$$

The string is $aaabbb$ is valid using derivation

$$S \rightarrow aSb | \epsilon$$

$$S \Rightarrow aSb$$

$$\Rightarrow aaSbb$$

$$\Rightarrow aaasbbb$$

$$\Rightarrow aaasbbb$$

$$\Rightarrow aaabbbb$$

$\therefore 'aaabbbb'$ is valid

10. Convert the following ambiguous grammar
 to an unambiguous grammar:

$$E \rightarrow E + E \mid E * E \mid Id$$

$$E \rightarrow E \wedge T \mid T$$

$$T \rightarrow T \wedge F \mid F$$

$$\boxed{F \rightarrow Id}$$

This ensures that multiplication has higher precedence
 than addition.

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow Id$$

$$E \rightarrow E \wedge T$$

$$T \rightarrow T \wedge F$$

$$F \rightarrow Id$$

$$Id \rightarrow Id$$

$$Id \rightarrow Id$$